SPR Optimizer: Results & Analysis

This report details the findings of the Surface Plasmon Resonance (SPR) optimization simulation. It includes an analysis of the convergence history, parameter sensitivity, and the optimal design parameters found.

1. Optimization Convergence History

The graph below tracks the optimization process over 20,000 trials. The Y-axis represents the objective value (negative Figure of Merit), meaning lower values are better.

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Analysis of Figure 1:
The red line indicates the best value found so far. As observed, the optimizer converges extremely rapidly, finding a near-optimal solution (FOM ~187) within the first 50-100 trials. The subsequent flat red line indicates that despite thousands of further trials (represented by the blue dots), no significantly better design was found.

The scatter of blue dots shows the algorithm exploring different regions of the parameter space. The presence of positive values (blue dots above 0) indicates sensor configurations where the resonance shift was inverted or non-existent, resulting in a failed objective score.

2. Hyperparameter Sensitivity Analysis

This analysis determines which physical parameters have the greatest impact on the sensor's performance.

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Analysis of Figure 2:
The bar chart quantifies the importance of each design parameter:
1. **D3_nm (Dielectric Layer 1):** This is the most critical parameter with an importance score of ~0.43. This suggests that the thickness of the $PbMoO_4$ layer is the primary driver of the resonance condition.
2. **D2_nm (Metal Thickness):** The second most important factor (~0.30). This controls the coupling of light into surface plasmons.
3. **Metal Choice:** Surprisingly, the choice of metal (Ag vs Au vs Cu) is less critical (0.13) than the geometry of the layers.

Conclusion: Precise fabrication control of the dielectric layer thickness (D3) is more important for this specific sensor design than the choice of metal.

3. Optimal Design Parameters

Based on the simulation data, the highest performing configuration found was:

Performance:
- Figure of Merit (FOM): **187.06**

Physical Structure:
- **Metal:** Gold (Au)
- **Metal Thickness (D2):** 24.4 nm
- **Dielectric 1 (D3):** 13.5 nm
- **Dielectric 2 (D4):** 1.6 nm